



distribute it when needed.

Both Ashwin and Shreyas think that the demand for new-age batteries will be in the area of aviation for applications like drones and further in smart-city technologies, most importantly for grid storage. But according to Savio, "If power cuts are high (say, more than four hours), battery storage would become a difficult proposition as the current storage has only 4-hour backup; hence, diesel generator (DG) sets still continue to be a viable option."

Talking about evolution of energy storage requirements in commercial and industrial sectors, the economics of using Li-ion batteries or advanced lead-acid batteries for behind the meter applications does not make much sense. The DG sets are much cheaper and more reliable than battery packs. As we scale up the supply of batteries, we see that changing and, in the next 3-5 years, the economics of it could start favouring their usage behind the meter energy storage.

Swappable batteries

EVs survive on batteries and are hence inseparable from each other. But unlike yesteryears, when EVs had only fixed battery systems, swappable batteries are giving competition to their non-removable counterparts. Ashwin explains, a fixed battery

system is one where the battery is embedded in the chassis of the vehicle and is non-removable, whereas a swappable battery is one that can be removed from the vehicle and replaced with a charged battery.

There are of course some differences in vehicles designed for fixed vs swappable batteries. In a fixed battery system, the chassis is designed for a fixed battery. In a vehicle designed for swappable batteries, the batteries have to be easily removable, and must be able to withstand mechanical vibrations. Also, in a swappable battery system, the batteries connect and disconnect frequently, and therefore the connectors have to be designed accordingly.

Indian 2-wheeler manufacturer Hero MotoCorp is partnering with Gogoro, a pioneer in swappable battery systems in the 2-wheeler space. But firms like Ola Electric and Aether have made their bets on fixed battery systems. According to Ashwin, the concept of fixed and swappable batteries is a little nuanced. In practical cases, where the usage of the vehicle is not so extensive, a fixed battery would be a better option as the vehicle is charged every night or while in parking. In cases where the vehicle travels 80-100 kilometers a day, where recharging is not convenient, a swappable battery may be needed.

Such vehicles include commer-

cial fleets such as e-rickshaws or last mile delivery vehicles. Any downtime in such cases creates a loss of significant revenue. In nutshell, private vehicles could look at fixed batteries whereas the commercial fleets may require swappable batteries so they do not waste productive time in recharging.

Shreyas thinks that it is not one size fits all with batteries and mobility. When it comes to swappable batteries, their faster turnaround time is very advantageous. Moreover, it also allows the economy to play around different models of ownership of the battery, bringing down the cost of ownership of vehicles.

However, there are challenges and hurdles in the swappable batteries model as well. The concept of swappable batteries has emerged in the last couple of years. Though it is preferable in fleet and commercial workspace, challenges with respect to standardisation, energy specifications, form factors, connectors, etc persist.

As per Ashwin, a major challenge faced is in the case of commercial vehicles. Their operators do not wish to invest on the batteries. But financing in this area has still not been figured out.

As regards battery swapping in private 2-wheelers, the economics do not make sense as the battery prices are falling. Hence, for personal vehicles, fixed batteries are relevant.

Shreyas points out that the problem of standardisation and interoperability is a regulatory issue. For swappable batteries, the battery has to be compatible with the vehicle used, which is also a roadblock at present.

Fast charging infrastructures are growing rapidly and there is a decrease in the need for swappable batteries (at least in private vehicles). But in the commercial space, according to Shreyas, from an affordability perspective, the fast charging technology is not there yet.

The stations may become more energy efficient and fast, but another challenge of fast charging stations (or any charging stations) is their dependence on the power grid itself, which is unreliable in certain parts of the country. This is a larger infra-